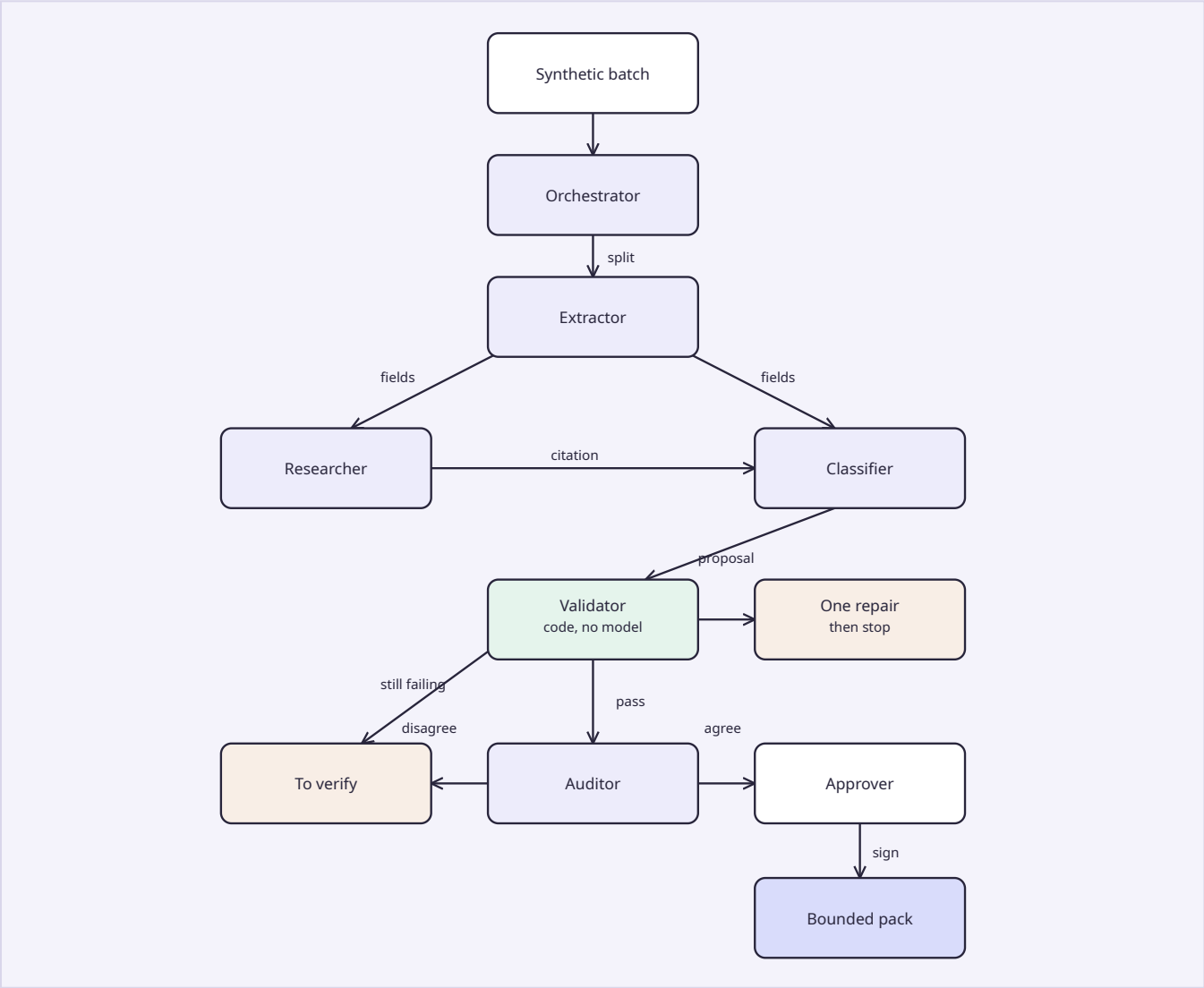
Explanation

How a review runs, which part of the assignment it answers, and which tool does the work.

Desert Ledger Trading, on the desk as a synthetic tenant. Research date 7 November 2026. In Arabic this page is labelled الشرح. Diagram labels stay in English and run left to right.

How a review runs

Someone loads a synthetic batch for one tenant. The orchestrator splits that batch. The extractor writes the fields and keeps the page each field came from. The classifier proposes a VAT treatment and adds a labelled note on corporate tax. The researcher may attach only a passage from the frozen FTA and MoF register, and that passage has to carry its version. The validator is Python. It checks the schema, the required fields, and the arithmetic, and it can send an invoice back once. A missing buyer identifier stays "to verify", because the repair cannot invent one. The auditor samples the call without the classifier's reasoning. A disagreement is escalated. The person approves, rejects, or escalates. What they sign is a readiness note on this desk.



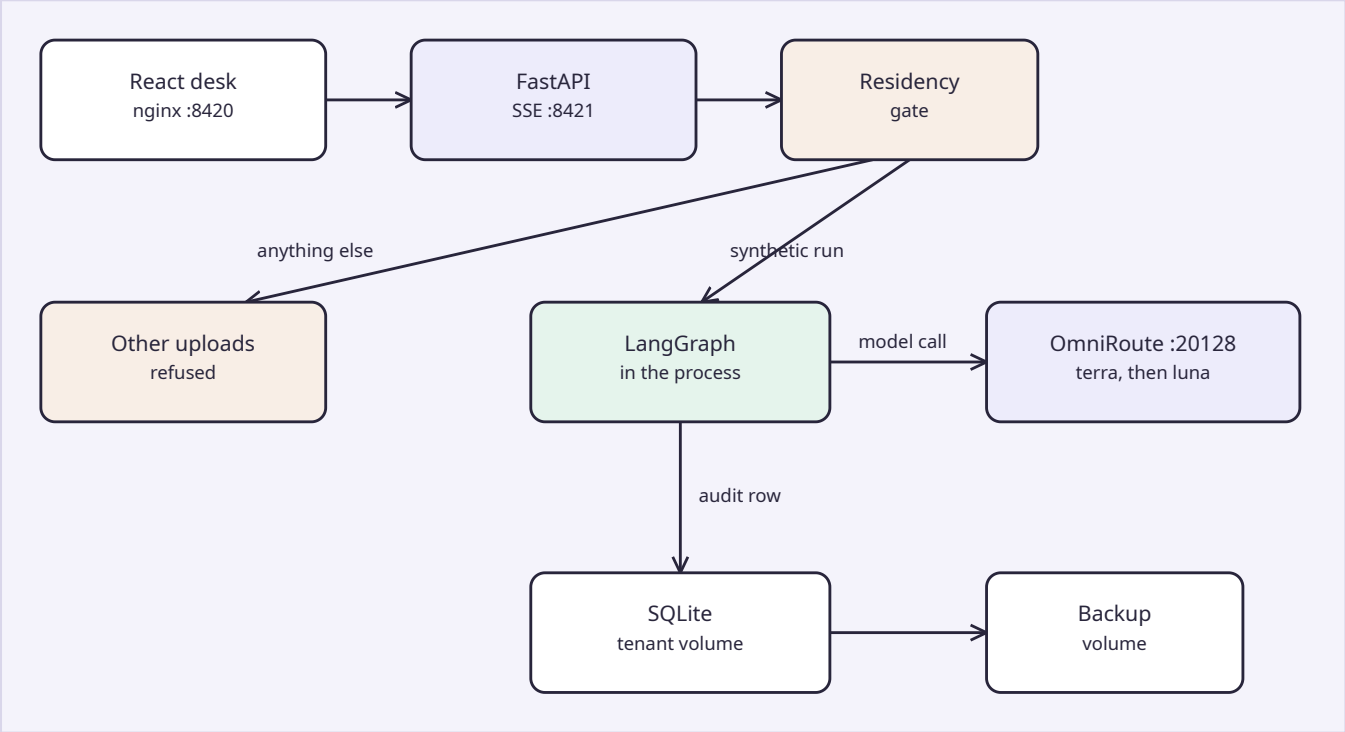
Lavender nodes call a model. The green validator does not. A sand coloured box is a stop: one repair, then the row is held, or the auditor disagrees. The approver sees both.

Where the work sits

The desk is React and Vite, served by nginx on port 8420. The API is FastAPI on port 8421. Each node emits a server-sent event, so the trace on screen is the run. SQLite on a tenant volume stores the document hash, the source version, the model, the checker outcome, and the person who signed. A second volume holds the backups.

Every model call is one chat completion to the local OmniRoute gateway on port 20128. The primary model is `gh/gpt-5.6-terra`. One retry uses `gh/gpt-5.6-luna`. This repository does not start that gateway and does not change it. If the gateway is down, the desk plays a labelled replay and says those timings are not live measurements.

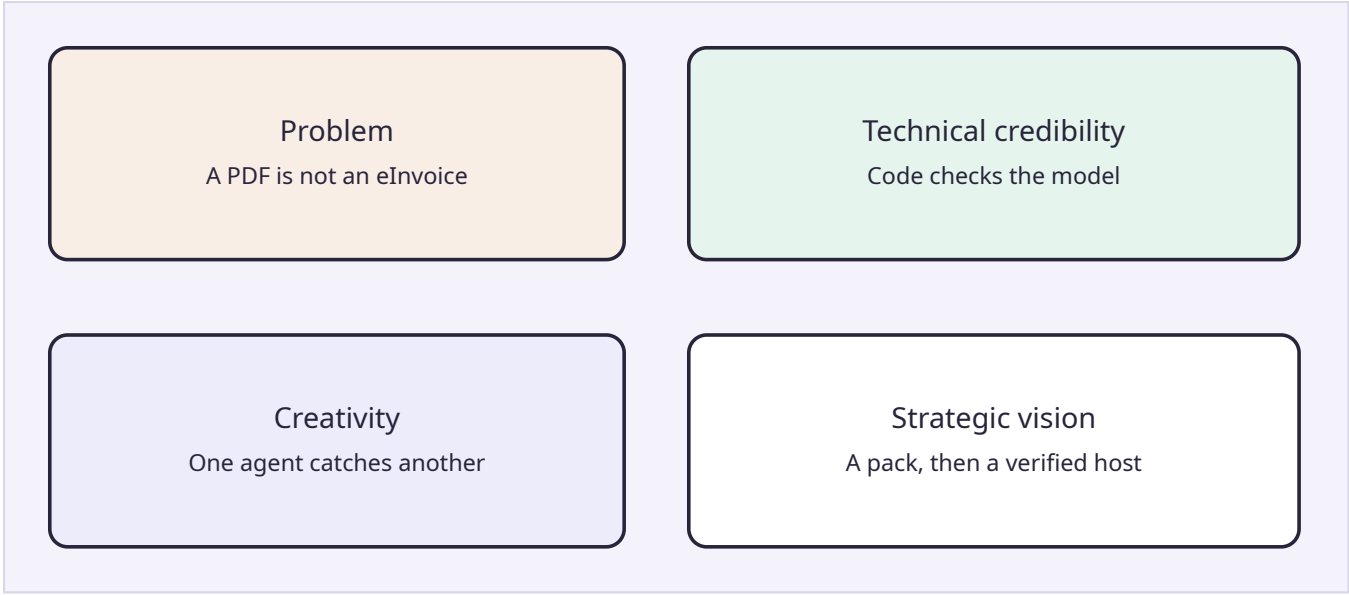
`RESIDENCY_VERIFIED` defaults to false. A localhost address does not show where an upstream provider processes the data. While the flag is false, only synthetic content is sent. Any other upload is refused before a model call.



The gate sits in front of the graph. A synthetic run may call OmniRoute. Anything else stops. The audit row is written to the tenant volume and copied to the backup volume.

How this answers the assignment

The brief asks for a functioning code-first agent, a complete frontend and backend, and a 15-minute presentation: the problem, the architecture and tools, a live demo, the challenges, and a DataRobot vision.



Problem

UAE finance teams cannot treat a PDF as an eInvoice. The required invoice is structured data. The original July 2026 appointment date moved, and the FTA notice of 25 October 2026 schedules the larger cohort at AED 50 million and above. One model checking its own batch would hide a bad total or a stale deadline. The work is split so each role owns one of those risks. The research date is 2026-11-07. The tenant on the desk is Desert Ledger Trading, and the registration number is labelled synthetic.

Technical credibility

Each role is a LangGraph node with its own prompt and a Pydantic output schema. The validator is ordinary code. Repair stops after one attempt. The auditor does not receive the classifier's reasoning. A cross-tenant read is refused before lookup. Approval waits until the checker has finished and the document hash is unchanged. The supported start is `docker compose up`. A reviewer does not install Python or Node on the host.

Creativity

Invoice DL-INV-1002 is missing a buyer identifier, and its totals do not meet. The classifier still proposes a treatment. The validator names the failed rule, uses the one repair, and holds the row. The auditor disagrees on the free-zone invoice and on the conflicting revenue invoice. An older deadline loses to the newer FTA source. Asking in Arabic why AED 375,000 does not set the eInvoicing phase returns a desk translation with the English passage beside it. That figure is a corporate tax band.

Strategic vision

A customer would keep this agent core and swap a versioned jurisdiction pack, then run it in containers with storage, logs, backups, and inference pointed at a location that has been checked. The build shows that gate. It leaves OmniRoute unverified. No savings figure is claimed, and nothing is sent to an accredited provider.

What the six demo minutes show

The tenant and its stated revenue, the batch splitting on the trace, the seeded rejection, the Arabic threshold question with the English citation, the auditor's disagreement, and a bounded signature. A refusal is part of the same path. Challenges a speaker can say are the ones written after the build: the AED 50 million sources disagree on "equal", the repair cannot invent a buyer, and the 90 percent accountant target is not a result.

Tools used to deliver it

The same path as the runtime diagram, with the reason each tool is in the repository.

Tool	What it does here	Why it is in the repository
LangGraph	Runs the seven roles as nodes in one process.	The handoff is what a reviewer can watch. No graph database is called.
Pydantic	Schema for each node's output.	The next node has to take those fields. A free paragraph is not valid output.
Python validator	Buyer identifier, totals, duplicate id, stale deadline, and the AED 50 million boundary.	Ordinary Python does these checks, and the demo uses them as its proof.
FastAPI	Documents, runs, server-sent events, decisions, reports, and questions.	The desk and the graph share one contract.
React and Vite	Companies, statistics, the review, and this explanation.	The reviewer watches the trace, asks, and signs.
SQLite	Tenant, document hash, source version, model, checker, and approver.	The audit row stays on the volume after a restart.
Docker Compose	A frontend container, an API container, and two volumes.	The start command is docker compose up.
OmniRoute	OpenAI-compatible chat completions on port 20128.	One local gateway. This repository does not call a provider SDK.
pytest	Arithmetic, abstention, cross-tenant reads, thresholds, the stale deadline, and the residency refusal.	The demo claims are tied to an answer key.
Gitleaks, TruffleHog, Trivy, Semgrep	Scan this repository from the security profile.	Secret verification stays off.